

Céspedes.Agro



What I am doing and why

Céspedes.Agro is an intelligent experiment I develop in my free time (hobby). It is not a commercial project, but a proof of concept that demonstrates that urban farming can be managed with accessible technology and low budget.

I have achieved that sensors, actuators and a web interface "talk" to each other. From any device (mobile, tablet or computer) I can monitor climate, irrigation and automation in real time. Although it is in development, the architecture and data flow already work end-to-end.

This IoT project lays the foundations for, in the future, incorporating industrial sensors and artificial intelligence to help plan crops.



From concept to working hardware

The project went from an idea on paper to two physical enclosures with real components, working stably 24/7.

Box 1: The brain (Orange Pi Zero 3) Inside this box is the central server. An Orange Pi Zero 3 running GNU/Linux Debian 12 that runs:

- Mosquitto (MQTT broker) that receives all sensor data from the field node.
- Nginx that serves a web page accessible from any device on the network.
 - ✓ The web interface receives the sensor data from the node via a WiFi router that I tested with coverage both in the growing area and in the server area, showing in real time temperature, humidity, light, flow rate, water temperature and rain detection. Moreover, from that same page the farmer can turn the irrigation solenoid valve and the humidifier on or off.

Box 2: The field node (ESP32 CP2102) In the second box is the ESP32 with all sensors and actuators soldered. Its job is:

- Read ambient temperature and humidity (HDC1080).
- Measure water temperature (DS18B20).
- Measure light intensity (BH1750).
- Count water flow (YF-S201).
- Detect rain (FC-28).
- Send everything via WiFi to the Orange Pi MQTT broker.
- Receive orders from the web, allowing the farmer to activate or deactivate the solenoid valve and humidifier relays as needed.

- Bluetooth backup layer:
 - ✓ The ESP32 also transmits the same data via Bluetooth to a mobile phone, without relying on the WiFi network. This method allows knowing the IP address assigned to the ESP32 and the IP that the farmer must use to monitor the web interface. Popular applications like BT Terminal or Bluetooth Terminal are suitable for receiving the data from the node.

The result

The data travel the complete path:

- Physical sensor → ESP32 → WiFi → MQTT → Orange Pi → web interface → browser.
- And the farmer's commands travel the opposite direction to the actuator (relays / solenoid valve). That is the basis of a smart farming system, and it already works here.

What is missing (future)

This is not finished. It is a solid foundation to build upon:

- Add soil moisture sensors (HW-390) for finer irrigation monitoring and control. Although these sensors are more experimental and educational, they could be used to train an AI on how quickly crops consume water from a container based on light intensity, crop type, amount of soil and ambient temperature. However, deciding which microcontroller to use in a second node to deploy the HW-390 has become a critical point, since the Arduino Nano I own (which can perfectly handle 7 HW-390

simultaneously with stable 5 V inputs and outputs) has no WiFi module, and the popular ESP01 modules are not reliable for 24/7 transmission.

- Centralised energy and lighting control using the Orange Pi GPIO pins. For lighting, I find it interesting to use full-spectrum (growth) LED strips to accurately simulate the circadian cycle — sunrise, noon and sunset — using PWM to gradually adjust light intensity. This dynamic control will not only optimise crop biological development, but will also be essential for generating a rich dataset to train an AI on the correlation between light intensity, temperature and water consumption rate recorded by the moisture sensors.
- Improve the web crop planner so that daily water is automatically adjusted according to the real phase of each plant (currently only potatoes have a progressive curve; the rest use constant irrigation).
- Incorporate a small artificial intelligence that analyses historical data and advises the farmer when to sow each crop according to the recorded environmental conditions.

